Option C: Malware / Tool Analysis (Malware of the Day)

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AsyncRAT: Remote Access Trojan Analysis

1. Malware Overview

Attribute	Details
Malware Name	AsyncRAT
Type	Remote Access Trojan (RAT) / Remote Access Tool
Platform	Windows
Primary Purpose	To provide attackers with Unauthorized remote access and control over an infected windows system.
MITRE ATTECK ID	S1087
Main Risk	Remote control, surveillance, data collection, and unauthorized access

Description:-

AsyncRAT is an open-source remote access tool that has been used in malicious campaigns. Although remote-access software can have legitimate administrative uses,

unauthorized deployment of AsyncRAT can provide attackers with control over compromised Windows systems.

2. Infection Chain

AsyncRAT can infect a system through phishing emails, malicious attachments, malicious links, fake software, and other social engineering techniques. In a typical infection chain, the victim receives a malicious file or link and executes it. The malware then runs on the system and establishes communication with its Command and Control (C2) server.

A common infection chain is:

Phishing Email / Malicious Link

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Victim Opens or Executes the file

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AsyncRAT Payload Executes

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RAT Establishes Persistence

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System Connects to Attacker-Controlled C2

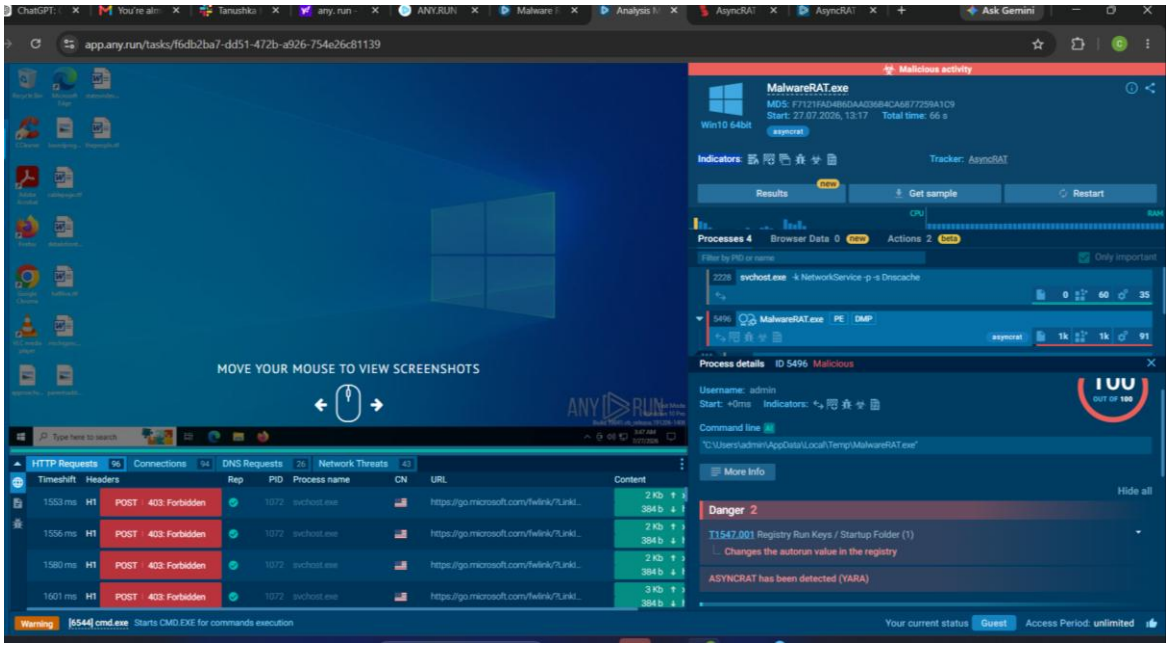
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Attacker Performs Remote Activities

3.Persistence

AsyncRAT can maintain persistence on a compromised Windows system by using mechanisms such as Scheduled Tasks. A scheduled task can be configured to automatically execute the malware at system startup or when a specific trigger occurs, allowing the malware to remain active even after the system is restarted.

MITRE ATT&CK Technique: T1053.005 – Scheduled Task/Job: Scheduled Task



Persistence:

The analyzed AsyncRAT sample modified an autorun value in the Windows Registry to maintain persistence. This behavior maps to MITRE ATT&CK T1547.001 – Registry Run Keys / Startup Folder.

Evidence: Any.Run analysis detected a change to an autorun value in the Registry.

4.Command &Control (C2)

The analyzed AsyncRAT sample generated significant network activity, including HTTP requests, network connections, and DNS requests. The sandbox recorded 96 HTTP requests, 94 connections, and 26 DNS requests during the analysis. This demonstrates that the sample communicated over the network; however, the exact malicious C2 domain/IP s hould be identified from the specific network events attributed to the AsyncRAT process.

Observed Network Activity:

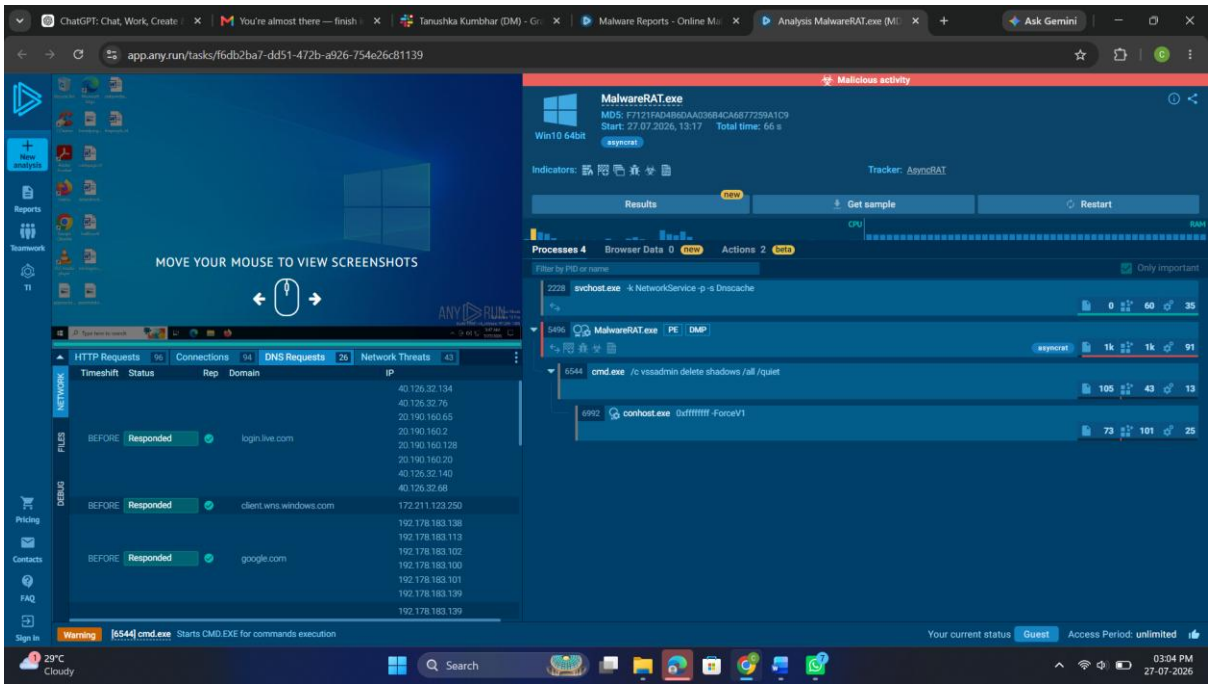
HTTP Requests: 96

Connections: 94

DNS Requests: 26

Network Threats: 43

Purpose: Command execution and communication with the infected system



5. Capabilities

AsyncRAT is capable of performing several remote-access and surveillance activities on a compromised Windows system. Its capabilities may include executing commands, collecting system and user information, monitoring processes, capturing keystrokes, taking screenshots, and transferring files between the infected system and the attacker.

Main Capabilities:

- Command Execution: Executes commands on the compromised system.
- Keylogging: Captures user keystrokes.
- Screen Capture: Captures screenshots of the victim's screen.
- System Information Collection: Collects information about the computer and operating system.
- Process Discovery: Lists running processes.
- File Transfer: Can download or transfer files.
- Remote Access: Allows unauthorized interaction with the compromised system.

6. Anti-Analysis

The analyzed AsyncRAT sample may use anti-analysis techniques to make detection and analysis more difficult. Potential techniques include obfuscation, virtual machine

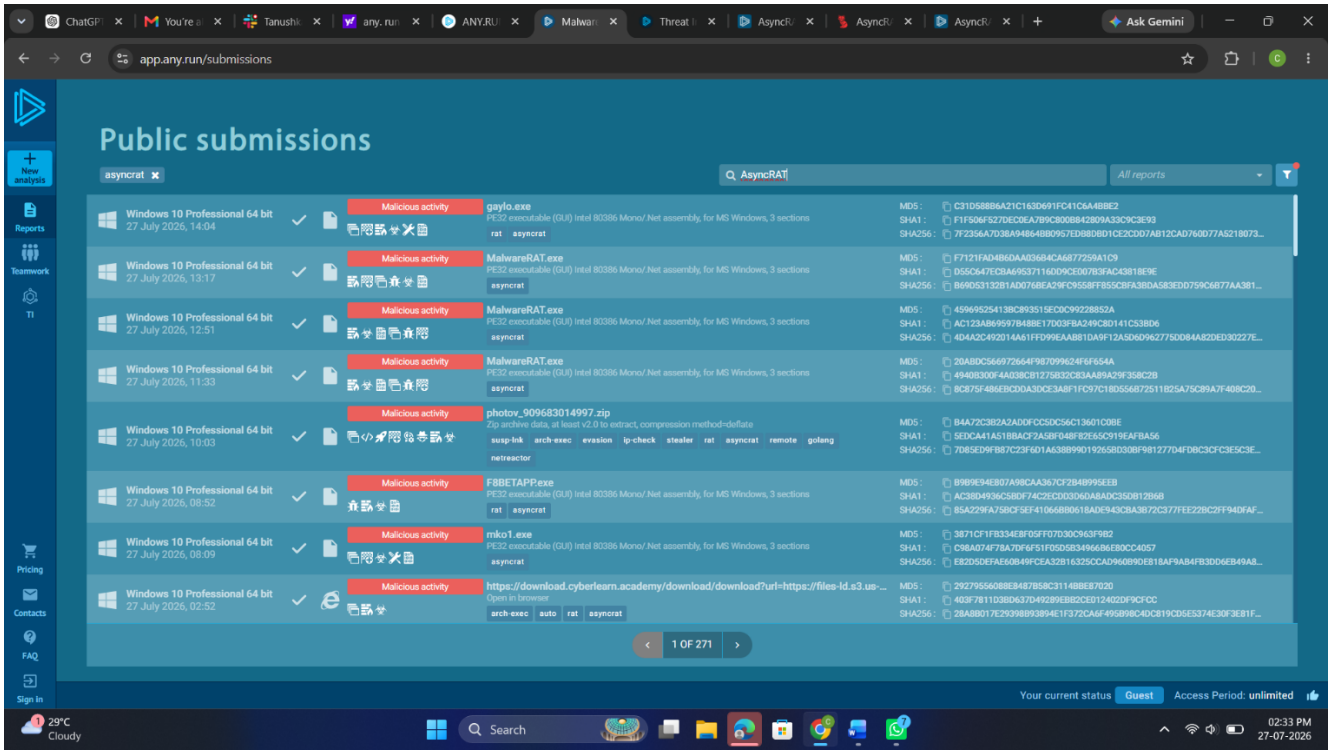
(VM) detection, sandbox checks, and encrypted configuration data. These techniques can help the malware evade automated analysis environments and hinder security researchers from understanding its behavior.

7. MITRE ATTECK Mapping

Observed Behaviour	MITRE ATT&CK Technique
Registry autorun value modified for persistence	T1547.001 – Registry Run Keys / Startup Folder
Commands executed using cmd.exe	T1059.003 – Windows Command Shell
System information collected	T1082 – System Information Discovery
Running processes discovered	T1057 – Process Discovery
Screenshots captured	T1113 – Screen Capture
Files transferred	T1105 – Ingress Tool Transfer

8. Screenshots

1. AsyncRAT sample analyzed in the Any.Run interactive malware sandbox.



9. Detection

AsyncRAT can be detected using a combination of YARA rules, endpoint monitoring, and network-based detection. Security teams should monitor suspicious Registry Run Key modifications, unexpected command execution through cmd.exe, unusual outbound network connections, and suspicious DNS activity. YARA rules can also be used to identify known AsyncRAT files and malware characteristics.

Detection Guidance:

- Monitor suspicious Registry Run Key modifications.
- Monitor unusual execution of cmd.exe by unknown applications.

- Detect suspicious outbound network connections.
- Monitor unusual DNS requests and C2 communication.
- Use updated antivirus, EDR, and threat intelligence indicators.
- Use YARA rules to identify known AsyncRAT samples.

10. References:-

Malware analysis Details:-https://any.run/malware-trends/asyncrat/

Daily report:- https://www.yazoul.net/malware/async-rat/

Sandbox analysis :- https://app.any.run/submissions

C2/network activity :-https://app.any.run/tasks/f6db2ba7-dd51-472b-a926-754e26c81139